Lecture 1.1 – Introduction

Working for 'South Land Engineering' – an *totally real* engineering firm.

Project

- Develop a prototype
- Write and present a business proposal
- Multi-disciplinary
- Senior staff member as a mentor
- Technical support forum (on Slack)
- *Details are in manual*

Project Coordinators

- Craig Sutherland (cj.sutherland@auckland.ac.nz & capstone subject) – 405-739
- James Tizard (james.tizard@auckland.ac.nz)

Project Deliverables

Risk Analysis	8 March
Project Proposal	28 March
Proposal Peer Reviews	19 April
Final Report	24 May
Personal Reflection	31 May

Meeting with Mentor

- Every week
- Answer

Guest Lecture – Student Group Projects: Challenges and Risks

1. Risk
2. Challenge
3. Challenge
4. Risk
5. Risk
6. Challenge
7. Risk

Nice all right

Risks in student projects:

- Student contribution
  - Engagement: Interest
  - Expertise
  - Personality
  - Availability
  - Wellbeing
- Team self-management
  - Communication

  - Coordination
  - Cooperation
  - Process
  - Clarity
- Resources
  - Hardware
  - Software
  - *Technology*

Risk mitigation

Lecture 2: Te Tiriti o Waitangi

- Signed in 6 Feb 1840

Two versions

- Maaori
- English

Three articles:

1. Governance
2. Full Rights of Chieftainship
3. Protection

From these articles three treaty principles are derived:

- Partnership
- Participation
- Protection

Maaori Society

- Iwi
- Hapuu
- Marae
- Whaanau

Te Ao Maaori

- *Tikanga*: customary lore and beliefs
- Differences on concept of ownership

Data Soverignty

- Whose data?
- Whose ethics?
- Whose decisions?

Indigenous Data Sovereignty

- Data from Maaori
- Data about Maaori
- Data about Maaori resources

Maaori Data Sovereignty:

> Maaori Data Sovereignty refers to the inherent rights and interests Maaori, whaanau, hapuu, iwi and Maaori organisations, have in relation to the creation, collection, access, analysis, interpretation, management, dissemination, re-use, and control of data relating to Maaori, whaanau, hapuu, iwi and Maaori organisations as guaranteed in Article II of Te Tiriti o Waitangi
> — Taiuru, K. 2020

For us:

- Ethnicity, Iwi of Children
- Clinics, Marae, Public sites
- Cloud storage, might not be stored in New Zealand
- Which groups outside of Plunket would need access?

Co-design

- Collaborative Process
- User Centered
- Iterative and Adaptive
- Empathy and Understanding
- Shared Ownership

what *noa*

User Requirements: Plunket & Weighing Babies

Plunket nurse: Christie

Plunket Overview

Plunket is a charity helping families with healthcare with their young children

Goals:
- Mauri ora: healthy babies and children
- Whaanau ora: healthy confident families
- Wai ora: healthy environments and connected communities

Healthcare Staff
- 523 frontline nurses
- 90 community Karitaane (health workers)

Plunket sees:
- 45k babies a year
- 280k children a year

Plunket Nurses

Visits from 0-5 years old

Core Visits include growth checks such as weight and are done at:
1. 4-6 weeks
2. 8-10 weeks
3. 3-4 months
4. 5-7 months
5. 9-12 months
6. 15-18 months
7. 2-3 years old
8. 4 years old

Where visits occur:
- Home visits
- Clinic visits
- Community
- Daycare

Visit components:
- Parental health (mental, physical)
- Family health (financial)
- Child health (including behaviour)
- Development (language and motor skills)
- Physical Assessment (shape, motor & sensory function, teeth)
  Measurements:
  - Head circumference
  - Height
  - Weight

Weighing Babies

- Scales have lip to prevent baby from rolling out
- Calibration every 6-12 months
- Infection control: disinfected, clean poo, pee etc.
- Cover tray: softer surface and protection from cold and protect scale
- Take naked weight of baby
- record measurements in ePHR (internal plunket system) to 0.01 kg precision (user input errors)
- Portability: scales need to be portable for going to house visits etc.
- Current scales only go to 20 kg (older babies sometimes weigh more than this)
  - possibly more precise scale?
  - more than 20 kg would be good!
- Some scales have tub and scale to stand on
- Scales take an average and settle on a number eventually
- Sometimes adult may hold child to weigh and stand on scale together if child is too restless

Growth chart

Weight data goes into growth chart with lines from data from WHO on typical baby development to compare against the babies development (line chart)

Mainly important for premature babies, age may be offset to reflect this

Down syndrome is a significant factor for growth and is not currently represented in growth chart application (typical growth charts are available)

Will be shown to parent

Export as PDF

Procedure

- Wipe down scale
- Cover tray & remove clothes from baby
- Turn on scale and zero
- Weigh baby
- Record reading

Other

Have to take notes on babies anyway so laptop okay. Have used tablets in the past

In system under NHI number and name.

Nutrition is relevant to weight and so data is recorded on it (maybe include???)

Sometimes do not have internet connection. (wifi bad lmao)

Birth weight also in record

Parent access: sometimes parents enter details as well?

Guest Lecture: Design for Sustainability

Introduction to Sustainable Development

Technical
- Limited quantity of resources
- Earth's resiliency

Ethical
- Equity

Chairs

Silly Cardboard Chair
- cardboard grows on trees!
- decays!
- low durability

Savonarola Chair

- high durability
- higher material cost

Weird office chair
- good because leasing model systemic motivation for long lasting chair

Beyond Material Sustainability

Life cycle perspective
1. Use
2. Recycle
3. Resources
4. Manufacturing
5. Assembly
6. Retail
7. And start from 1

Foundations for Sustainable Change – Iceberg
1. Technical Innovation ("doing")
2. Social Innovation ("being + doing")
3. Cultural change ("thinking")

Global Observation

- Ecocentrism
- Suboptimality can be desired for error tolerance.
- modular, decentralised and diverse systems are intrinsically error friendly